

meaning that, among other things, we examined historical data outside the period tested by Greenblatt and controlled for small-capitalization stocks. Our findings surprised us, and revealed a deep truth about value investment, one that is often lost in the scramble to emulate Buffett.

ANALYSIS OF THE MAGIC FORMULA

Our independent investigation into the Magic Formula found that it did outperform the market, and by a wide margin, as Greenblatt claimed, but not in the same magnitude. We also found that the returns were concentrated in smaller market capitalization stocks, but not so small as to be uninvestable for all but the very largest institutional investors. We examined the performance of the Magic Formula over the period 1964 to 2011—considerably longer than the period examined by Greenblatt, and including years both prior to and after his study—and excluded smaller capitalization stocks, which we defined as any stock falling outside the S&P 500 Index. For context, the smallest company in the S&P 500 Index in January 2014 is \$3.4 billion, the median market capitalization is \$16.5 billion, and the average is \$35 billion. These are very large companies. We also weighted the remaining stocks by market capitalization to make the portfolios comparable to the S&P 500 Index, which is market-capitalization weighted, to adjust for any bias to small capitalization stocks. We found that, even under these arduous conditions, Greenblatt's Magic Formula outperformed the S&P 500 Total Return Index (which includes dividends, as does the Magic Formula) over the full sample, returning on average 13.9 percent annually compared to 10.5 percent for the market over the same period. Figure 4.1 shows a logarithmic chart of the performance of the market-weighted Magic Formula and the S&P 500 Total Return Index.

The Magic Formula achieved this 3.3 percent yearly outperformance at approximately the same level of risk (using the academic definition of standard deviation, 16.5 percent for the Magic Formula against 15.2 percent for the market). 330 basis points—3.3 percent yearly—might not look like a big edge, but compounded over the full period, it meant that the Magic Formula earned almost six times the market's return. \$10,000 invested in the Magic Formula grew to \$12.2 million, while the same amount invested in the market grew to just \$2.1 million. The Magic Formula's performance was also quite reliable through the full period, beating the market in 85 percent of rolling 5-year periods, and 97 percent of rolling 10-year periods. We concluded that, while the results were not as impressive as the equal-weighted results claimed by Greenblatt, in the very large capitalization stocks we tested the market-capitalization-weighted Magic Formula